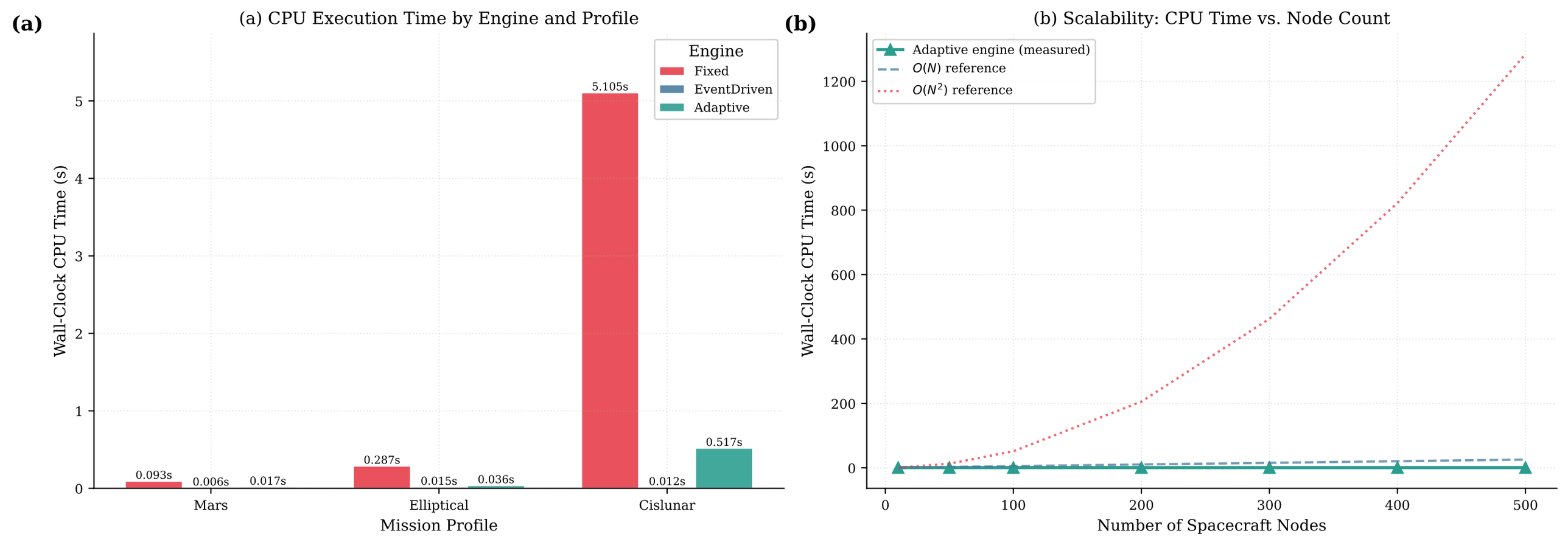
(a)
(a) CPU Execution Time by Engine and Profile
Engine
Fixed
EventDriven
Adaptive
Wall-Clock CPU Time (s)
0
1
2
3
4
5
0.093s
0.006s
0.017s
0.287s
0.015s
0.036s
5.105s
0.012s
0.517s
Mars
Elliptical
Cislunar
Mission Profile
(b)
(b) Scalability: CPU Time vs. Node Count
Adaptive engine (measured)
O(N) reference
O(N²) reference
Wall-Clock CPU Time (s)
0
200
400
600
800
1000
1200
0
100
200
300
400
500
Number of Spacecraft Nodes